

LAB 3: ARCHITECTURE JUSTIFICATION

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1. Architectural Choice

Layered Architecture is used for the Community Tool & Equipment Library System. The system is organized into a Presentation Layer for the Member and Custodian Portals, a Business Logic Layer for inventory, reservations, loans, payments and a Data Layer containing the Library Database. External payment and messaging services are accessed only through their responsible business components.

2. Two Reasons for Selecting Layered Architecture

- Clear separation of responsibilities: The user interfaces are separated from operational rules and data storage.
- Consistent rules and maintainability: All reservations, returns and inventory updates pass through the business layer. This provides one place to prevent duplicate reservations, enforce loan-duration limits and restrict overdue members from borrowing high-value equipment.

3. Security Advantage

Users cannot communicate directly with the database. The business layer can authenticate users, authorize member and custodian actions, validate inputs and record sensitive operations. Payment processing is isolated in the Payment and Fee Manager and delegated to the external Payment Gateway, reducing direct exposure of payment information to other components.

4. Performance Benefit

A layered application can process most requests through direct internal calls, avoiding the network overhead of a microservices architecture. Indexed database queries and caching in the Inventory Manager can provide fast equipment searches and availability checks while the centralized database maintains consistent data.